

26	B	Note that as the value of V is a straight line with a constant gradient, the square of this function will be a curve. Option D is wrong because it is obtained from an alternating function of constant V_0. Option C is wrong because it is obtained from an alternating sinusoidal function. Option A is wrong because it is the root-mean-square voltage is not the average voltage. Note: $V_{rms} = \sqrt{\frac{1}{T/2} \int_0^{T/2} V(t)^2 dt} = \sqrt{\frac{2}{T} \int_0^{T/2} \left(\frac{V_0}{T/2} t \right)^2 dt} = \sqrt{\frac{2}{T} \int_0^{T/2} \frac{4V_0^2}{T^2} t^2 dt}$ $= \sqrt{\frac{8V_0^2}{T^3} \left[\frac{t^3}{3} \right]_0^{T/2}} = \sqrt{\frac{8V_0^2}{T^3} \frac{T^3}{3 \times 8}}$ $= \sqrt{\frac{V_0^2}{3}} = \frac{V_0}{\sqrt{3}}$
27	A	Photoelectric equation: